重点速查：平均值与可合并状态

Mean aggregation / 中文理解 → English answers

一、中文理解

平均值不仅依赖各组的结果，也依赖组内记录数。任意局部平均值直接平均会改变观测的权重，因此中间结果必须保留总和与数量。

机制与前提

每个数值 x 变成 (x,1)，状态按分量相加。最后在 count>0 时计算 sum/count。局部结果要能继续作为同类型状态参与合并；数值溢出、浮点舍入和空组需要按任务约定处理。

手算反例

A=[2,4] 的均值为 3；B=[12] 的均值为 12。直接平均得到 7.5。正确结果为 (6+12)/(2+1)=6。若各组数量相同，直接平均局部均值才与正确加权结果一致。

易错点

局部均值丢失权重；中间状态不是最终答案；改变分区或合并树，不应在数学模型中改变业务问题。

二、考试英文表述

A mean cannot generally be combined by averaging local means, because the groups may have different counts. Preserve a sum-count state, merge both components, and divide only after the final aggregation. This retains each observation’ s weight. The state type must remain compatible with repeated merging, and zero counts and numeric division semantics need explicit handling.

Q1. Why is the unweighted mean of local means unsafe?

It treats groups as equally weighted even when their counts differ. For [2,4] and [12], it gives 7.5 instead of the global mean 6.

Q2. What information makes a mean safely mergeable?

Keep the sum and count of each partial group. Add the corresponding components and divide once at the end; this preserves the information required for correct weighting.